

Experiment No: 01

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Title: Different Data Types and Operators in Python

Program:

Taking inputs

```
num1 = int(input("Enter first integer: "))
```

```
num2 = int(input("Enter second integer: "))
```

```
float_num = float(input("Enter a float number: "))
```

```
text = input("Enter a string: ")
```

Data Types

```
print("\n--- DATA TYPES ---")
```

```
print("num1 type:", type(num1))
```

```
print("num2 type:", type(num2))
```

```
print("float_num type:", type(float_num))
```

```
print("text type:", type(text))
```

Arithmetic Operators

```
print("\n--- ARITHMETIC OPERATIONS ---")
```

```
print("Addition:", num1 + num2)
```

```
print("Subtraction:", num1 - num2)
```

```
print("Multiplication:", num1 * num2)
```

```
print("Division:", num1 / num2 if num2 != 0 else "Cannot divide by zero")
```

```
print("Modulus:", num1 % num2 if num2 != 0 else "Undefined")
```

Relational Operators

```
print("\n--- RELATIONAL OPERATORS ---")
```

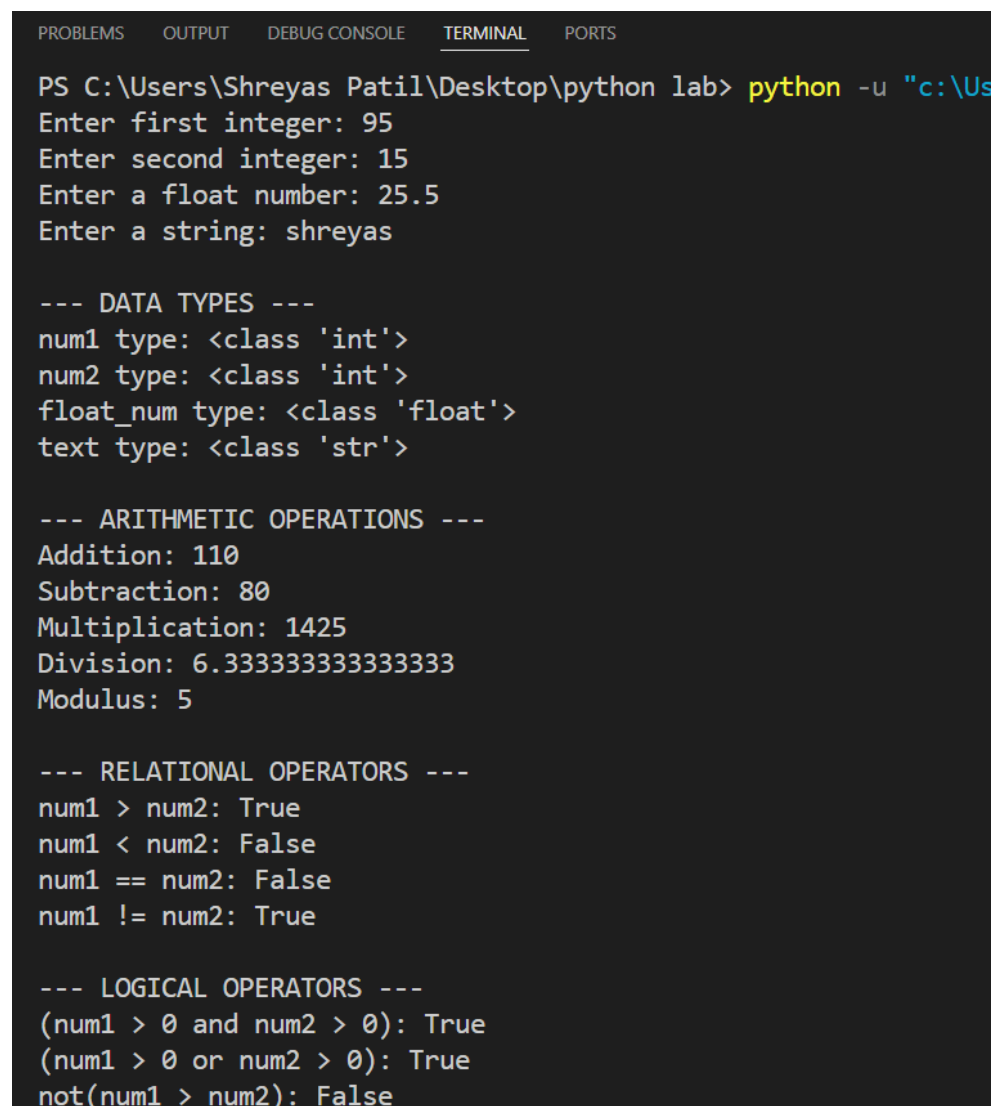
```
print("num1 > num2:", num1 > num2)
print("num1 < num2:", num1 < num2)
print("num1 == num2:", num1 == num2)
print("num1 != num2:", num1 != num2)

# Logical Operators

print("\n--- LOGICAL OPERATORS ---")

print("(num1 > 0 and num2 > 0):", num1 > 0 and num2 > 0)
print("(num1 > 0 or num2 > 0):", num1 > 0 or num2 > 0)
print("not(num1 > num2):", not(num1 > num2))
```

Output:



```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS C:\Users\Shreyas Patil\Desktop\python lab> python -u "c:\Users\Shreyas Patil\Desktop\python lab>
Enter first integer: 95
Enter second integer: 15
Enter a float number: 25.5
Enter a string: shreyas

--- DATA TYPES ---
num1 type: <class 'int'>
num2 type: <class 'int'>
float_num type: <class 'float'>
text type: <class 'str'>

--- ARITHMETIC OPERATIONS ---
Addition: 110
Subtraction: 80
Multiplication: 1425
Division: 6.333333333333333
Modulus: 5

--- RELATIONAL OPERATORS ---
num1 > num2: True
num1 < num2: False
num1 == num2: False
num1 != num2: True

--- LOGICAL OPERATORS ---
(num1 > 0 and num2 > 0): True
(num1 > 0 or num2 > 0): True
not(num1 > num2): False
```